

**U.S. Department of Justice
Drug Enforcement Administration**



**Temporary Placement of 7-Hydroxymitragynine Above a Specified Threshold into
Schedule I**

**Background, Data, and Analysis:
Three Factors Determinative of Control
and Findings Pursuant to 21 U.S.C 811(h)(3)**

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Introduction

7-Hydroxymitragynine¹ is a psychoactive alkaloid found in *Mitragyna speciosa* (*M. speciosa*) plant, a tropical evergreen tree indigenous to Southeast Asia. While the use of the plant was once geographically limited to Southeast Asian countries such as Thailand and Malaysia, the use of *M. speciosa* has since extended to the United States and other global markets. Among the numerous alkaloids identified in *M. speciosa*, the indole alkaloids mitragynine (major) and 7-hydroxymitragynine (minor) are primarily responsible for the plant's psychoactive effects. In its natural botanical form, 7-hydroxymitragynine makes up less than two percent of the total alkaloid content or occurs in trace amount in *M. speciosa*. However, 7-hydroxymitragynine can be synthesized from mitragynine through a one-step step chemical reaction and it also exists as an active oxidized metabolite of mitragynine *in vivo*. Despite their different origins, the chemical structures of synthetic and naturally occurring 7-hydroxymitragynine are identical (see Figure 1). Consequently, the intrinsic pharmacological profile, receptor affinity, and mechanism of action of 7-hydroxymitragynine molecule remain unchanged regardless of its source. While consumers of raw plant matrix may experience a modified or attenuated physiological effect due to the competitive, co-occurring alkaloids inherent to *M. speciosa*, isolated or semi-synthetically derived formulations deliver unattenuated, high-potency effects of the target alkaloid directly, representing a distinct public safety profile when concentrated above the proposed threshold.

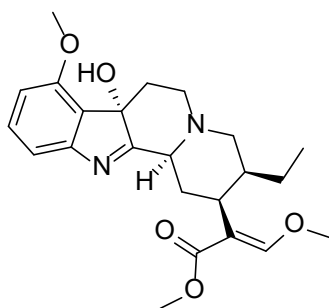


Figure 1: Chemical Structure of 7-hydroxymitragynine

Recently, the United States has seen a proliferation of 7-hydroxymitragynine products. Evidence suggests that commercially available products, including extracts and synthetic formulations, contain a significantly higher concentration of 7-hydroxymitragynine than what is found in botanical *M. speciosa*. These products are commonly sold on the Internet and in retail outlets, such as gas stations and smoke shops. They are distributed as individual dosage units, including pressed tablets, capsules, chewable, and gummies and, or as single retail containers, such as concentrated liquid bottles, vials, pouches, vape cartridges, and small extract jars. To date, the safety profile of

¹ Chemically known as methyl (*E*)-2-((2*S*,3*S*,7*aS*,12*bS*)-3-ethyl-7*a*-hydroxy-8-methoxy-1,2,3,4,6,7,7*a*,12*b*-octahydroindolo[2,3-*a*]quinolizin-2-yl)-3-methoxyacrylate (also known as: (α*E*,2*S*,3*S*,7*aS*,12*bS*)-3-ethyl-1,2,3,4,6,7,7*a*,12*b*-octahydro-7*a*-hydroxy-8-methoxy-α-(methoxymethylene)-indolo [2,3-*a*] quino lizine-2-acetic acid methyl ester)

these concentrated products in humans remains unknown because no controlled clinical trials have been conducted to establish safe consumption limits or standardized dosing

7-Hydroxymitragynine has opioidergic activity, sharing a similar pharmacological profile to schedule II opioids like morphine. Preclinical data indicates that 7-hydroxymitragynine carries a high abuse potential, with safety risks-including tolerance, dependence, and respiratory depression, which are comparable to those of classic opioid analgesics. The United States has recently seen an emergence of products containing 7-hydroxymitragynine, which are often characterized by ambiguous dosages and misleading marketing, frequently being labeled as “natural *M.speciosa* extracts.” While sellers promote these products for their psychoactive and opioidergic effects of only 7-hydroxymitragynine, evidence demonstrates these products may also contain other opioid alkaloids, such as mitragynine pseudoindoxyl. These combinations, coupled with a lack of regulatory oversight, pose significant safety risk to unsuspecting consumers by exposing them to high doses of opioids. The absence of clinical evidence to support vendor health claims is deeply concerning. Consequently, 7-hydroxymitragynine products pose significant health risks, as essential information regarding their purity, identity, quantity and long-term safety remain unknown.

7-Hydroxymitragynine does not meet the U.S. Food and Drug Administration’s (FDA) safety standard for drug, dietary supplements or dietary ingredients, and has not been proven safe or effective for any drug use. FDA has issued warning letters to companies clarifying that 7-hydroxymitragynine is not an FDA-approved drug product and no food additive regulation has authorized the use of 7-hydroxymitragynine in food supply. Furthermore, FDA has warned consumers against using products labeled as 7-hydroxymitragynine, as they have not been proven safe or effective for any use.² Fatal overdoses involving 7-hydroxymitragynine have been reported, making its wide availability and unknown safety profile a significant threat to public health, which is particularly concerning in the midst of an opioid crisis.

Background

By letter dated July 28, 2025, pursuant to 21 U.S.C. 811(b) and (c), HHS provided to DEA a scientific and medical evaluation entitled “Basis for the Recommendation to Control 7-Hydroxymitragynine and its Salts, As Present in excess of the Specified Threshold Limit Described Herein, in Schedule I of the Controlled Substances Act,” and recommended that 7-hydroxymitragynine as described below be permanently controlled in Schedule I of the CSA.

In a letter to HHS dated February 24, 2026, the Administrator of DEA notified the Assistant Secretary for Health (Assistant Secretary) of his intention to publish in the

² [FDA Issues Warning Letters to Firms Marketing Products Containing 7-Hydroxymitragynine | FDA](https://www.fda.gov/news-events/press-announcements/fda-issues-warning-letters-firms-marketing-products-containing-7-hydroxymitragynine#:~:text=FDA%20Issues%20Warning%20Letters%20to%20Firms%20Marketing%20Products%20Containing%207%2DHydroxymitragynine,-Alkaloid%20known%20as&text=The%20U.S.%20Food%20and%20Drug,also%20known%20as%207%2DOH), available at <https://www.fda.gov/news-events/press-announcements/fda-issues-warning-letters-firms-marketing-products-containing-7-hydroxymitragynine#:~:text=FDA%20Issues%20Warning%20Letters%20to%20Firms%20Marketing%20Products%20Containing%207%2DHydroxymitragynine,-Alkaloid%20known%20as&text=The%20U.S.%20Food%20and%20Drug,also%20known%20as%207%2DOH>,
H,

Federal Register a Notice of Intent to temporarily place 7-hydroxymitragynine above a specified threshold (derived from HHS's July 2025 scheduling recommendation)³ in schedule I of the CSA pursuant to the temporary scheduling provision of the CSA, 21 U.S.C. 811(h). The Assistant Secretary responded to this notice by letter dated March 6, 2026, and advised that based on a review by FDA, there are no approved new drug applications or investigational new drug applications for 7-hydroxymitragynine. The Assistant Secretary also stated that HHS has no objection to the temporary placement of 7-hydroxymitragynine above the specified threshold into schedule I of the CSA.

In light of the opioid pharmacological profile, false marketing claims, and associated risk with the misuse of 7-hydroxymitragynine, DEA reviewed the available scientific, medical, law enforcement, and all other relevant data on 7-hydroxymitragynine. With no approved medical use, the misuse of 7-hydroxymitragynine products underscores the public health threat posed by this substance. Thus, to protect public health and to avoid an imminent hazard to public safety, DEA intends to temporarily place 7-hydroxymitragynine above a specified threshold in schedule I of the CSA. The data presented in this document supports the determination by DEA to control this substance under the temporary scheduling provision of the CSA, 21 U.S.C. 811(h)(1). With respect to finding that control of this substance is necessary to avoid an imminent hazard to public safety, DEA has considered the three factors required under the CSA for the temporary scheduling of 7-hydroxymitragynine above a specified threshold. 21 U.S.C. 811(c)(4)–(6) and (h)(90).

Factor 4. Its history and current pattern of abuse

7-Hydroxymitragynine is most commonly marketed in isolated form or as component of *M. speciosa*. Historically, *M. speciosa* has been used for a variety of purposes including as an opium substitute and a treatment of various opioid withdrawal symptoms, such as pain, cough, anxiety, diarrhea, and intestinal difficulty (Cinosi et al., 2015; Dorman et al., 2015; Swogger et al., 2015; Garcia-Romeu et al., 2020; Heywood et al., 2024; Alsobrook et al., 2025; Grundmann et al., 2025). While the use of the plant was once geographically limited to Southeast Asia, the marketing in the United States of 7-hydroxymitragynine products has been aggressive and often indistinguishable from the sale of *M. speciosa* (Smith et al., 2025). In recent years, 7-hydroxymitragynine has been

³ The specified threshold for 7-hydroxymitragynine was adapted from the definition used by HHS, which is described as follows:

- (A) Any botanical material of the plant *Mitragyna speciosa*, also known as kratom, and contains more than 0.050 percentage of 7-hydroxymitragynine on a dry weight basis, or
- (B) Any alternative article or material to that described in (A), that is:
 - i. Resulting from synthetic methods and containing 7-hydroxymitragynine present in amounts greater than 0.050 percentage weight/weight, weight/volume, or volume/volume or greater than 1.00 milligram of 7-hydroxymitragynine in the article, or
 - ii. Material derived from *Mitragyna speciosa* and further processed to manufacture alternative dosage forms such as extracts, concentrates, processed edibles, or pressed pills, and which may have materials that have been exposed to chemical, thermal, or other methods leading to chemical transformations that result in 7-hydroxymitragynine present in amounts greater than 0.050 percentage weight/weight, weight/volume, or volume/volume, or greater than 1.00 milligram of 7-OH in the article.

unlawfully marketed as a natural extract or dietary supplement⁴ despite failing to meet FDA’s safety standard for dietary supplements or dietary ingredients. Anecdotal information from users who post on social media indicates that many users are self-treating chronic pain with unregulated 7-hydroxymitragynine products (NDEWS, 2025).

Product Formulation and Marketing Claims

7-Hydroxymitragynine products are readily obtained from smoke shops, gas stations, and online vendors in various forms, including tablets, liquid oral beverages (shots), capsules, powder, syrup, vapes, sublingual pouch/strips, nasal spray, chewable, and liquid extracts (Grundmann et al., 2025; Heywood *et al.*, 2024; Hill et al., 2025; Ogozalek 2023; Sharma, et al., 2024; Smith et al., 2025). These products are commonly sold with names such as “7 Ohmz”, “7-hydroxy”, and “7-OH” and they are presented in colorful packages. An investigation by Hill *et al* (2025) identified 250 products sold between September 2024 through February 2025, noting that chewable/sublingual tablets were the most common formulation. Sellers claim these products can be used for general wellbeing and increased focus. Specific marketing language for these chewable formats often emphasizes the ease of use:

“The chewable format is designed for convenience, allowing you to simply chew a tablet and savor the berry flavor. This eliminates the need for mixing or preparation... The compact size of each tablet makes them portable and perfect for on-the-go use, so you can carry them in your bag or pocket and take them whenever needed.”

Dosing and Cost Analysis

The Hill et al., investigation further highlighted the concerns regarding standardized dosing and cost. The concentration of 7-hydroxymitragynine products varied significantly, ranging from 1 mg - 700 mg in a single dose or serving. One example of a dosing recommendation states: *“Every pouch includes four expertly pressed classic extract tablets. These tablets are scored down the center for easy dosing and contain two servings total. Each serving is 7.5mg of pure 7-OH extract, for 15mg total per tablet.”* The average cost per dose across most products containing 7-hydroxymitragynine was about \$3.97 (Hill et al., 2025).

Table 1: Dosage and Cost of Some 7-hydroxymitragynine Products Sold Online

Characteristics	Minimum	Maximum	Mean
Total package (mg)	2	7000	120.48
Single Dose/Serving (mg)	2	750	36.41
Recommended Dose/Serving (mg)	1	700	21.92
Cost of Package (\$)	3.95	139.95	21.97
Cost/recommended Dose/Serving (\$)	1.14	17.99	4.04

⁴ FDA issued warning letters to firms marketing products containing 7-hydroxymitragynine stating it is not a lawful dietary supplement, food additive, or ingredient in any approved drug”. [FDA Issues Warning Letters to Firms Marketing Products Containing 7-Hydroxymitragynine | FDA](#)

Anecdotal information synthesized from user-contributed reports and subsequently analyzed by clinical educators between 2024 and early 2025 provide additional information on abuse patterns of kratom products. Users consistently report a transition from traditional kratom leaf to 7-hydroxymitragynine tablets (e.g., 7OHMZ, Press'd, Hydroxie). 7-Hydroxymitragynine was described as the “cleanest and most euphoric high” that lacks “ceiling effect of nausea” of the plain leaf. A dominant theme is the short-lived nature of the 7-hydroxymitragynine high. Users report the urge to redose frequently.

The National Drug Early Warning System (NDEWS) conducted web monitoring on reddit mentions of kratom and its derivatives. In the report, information provided by reddit discussants surrounding kratom and 7-hydroxymitragynine shows that users often compare 7-hydroxymitragynine effects to prescription opioids like oxycodone and hydrocodone, with users expressing worry on how such potent products are legally available at smoke shops. Discussants also mentioned the ease and convenience of buying products at gas stations and noted how this can be a concern for impulse use due to its ease of accessibility. Further, discussions on prices found that 7-hydroxymitragynine and kratom products ranged from \$15-40 per retail unit (NDEWS, 2025).

In summary, 7-hydroxymitragynine products are available in different forms and can be purchased easily via several avenues, predominantly via the Internet. This is a shift from traditional botanical use to aggressive marketing of highly concentrated products in the United States.

Factor 5: The scope, duration, and significance of abuse

The abuse of 7-hydroxymitragynine is increasing and widespread. Given its pharmacological profile as an opioid agonist with high abuse potential, the marketing of 7-hydroxymitragynine products to consumers as botanical supplements constitutes a significant public health risk.

Hill et al (2025) note that 7-hydroxymitragynine, a potent opioid, presents overdose risk and toxicity to users because its chemical manufacturing process is inconsistent and does not adhere to good manufacturing practices. This lack of oversight may lead to variation in doses and composition in final products. Further, 7-hydroxymitragynine is sold as sublingual pouch/tablet, a formulation known to deliver drugs rapidly into the systemic circulation. This rapid delivery, coupled with 7-hydroxymitragynine's known pharmacokinetic profile presents a significance of abuse (Hill et al., 2025; Maxwell et al., 2021; Sharma et al., 2025).

7-hydroxymitragynine sellers make claims of its opioidergic effects (Hill et al., 2025). Preclinical data indicates that 7-hydroxymitragynine carries a high abuse potential, with risks-including tolerance, dependence, and respiratory depression-comparable to those of classic opioid analgesics. Evidence demonstrates that 7-hydroxymitragynine products have been identified to contain other opioid alkaloids such as mitragynine pseudoindoxyl.⁵

⁵ [7-Hydroxy_Mitragynine_NPS_Discovery_033125.pdf](#)

These combinations and practice pose significant safety risk to unsuspecting consumers by exposing them to high doses of opioids.

Hence, 7-hydroxymitragynine products sold as unregulated dietary supplements pose significant health risks to users because information on their purity, identity, quantity and safety is unknown.

Law Enforcement Encounters

The number of seizures for 7-hydroxymitragynine as reported in law enforcement systems is currently limited, primarily because forensic chemists often prioritize the identification of mitragynine, the major alkaloid in *M. speciosa*, rather than extending analysis to the identification minor alkaloids such as 7-hydroxymitragynine. Furthermore, because 7-hydroxymitragynine is not federally controlled under the CSA, additional forensic identification may be limited. For example, some forensic laboratories may not place emphasis on analyzing or tracking the encounters of non-controlled substances, making it unlikely to be fully reported to forensic laboratories databases. Nonetheless, available data from the National Forensic Laboratory Information System,⁶ shows that there has been a total of 86 reports of 7-hydroxymitragynine from 17 states (see Table 2). In addition, according to the data report from the Department of Homeland Security (DHS) office of field operations, seizures of *M. speciosa* (identified as either mitragynine or 7-hydroxymitragynine) at ports of entry have fluctuated over the last decade, 27.18 kg in 2017, 0.52 kg in 2018, and 2,675.22 kg in 2019 (see Table 3).⁷ These law enforcement data collectively illustrate the widespread and increasing availability of products containing 7-hydroxymitragynine within the domestic kratom drug market.

Table 2: Summary of State Encounter of 7-hydroxymitragynine within NFLIS-Drug

State	2025*	2026*
Arkansas	12	4
California	1	0
Florida	6	10
Illinois	1	0
Indiana	6	1
Iowa	5	5
Louisiana	5	0
Maryland	0	1
Missouri	1	1
Nebraska	4	0
New York	1	0

⁶ The National Forensic Laboratory Information System (NFLIS) represents an important resource in monitoring illicit drug trafficking, including the diversion of legally manufactured pharmaceuticals into illegal markets. NFLIS is a comprehensive information system that includes data from forensic laboratories that handle more than 96% of an estimated 1.0 million distinct annual State and local drug analysis cases. NFLIS includes drug chemistry results from completed analyses only. While NFLIS data is not direct evidence of abuse, it can lead to an inference that a drug has been diverted and abused. See 76 FR 77330, 77332, Dec. 12, 2011. NFLIS data were queried May 21, 2026.

⁷ [Department of Homeland Security Border Security Metrics Report: 2021](#). Accessed 08/01/2025

North Carolina	1	1
Ohio	5	1
South Carolina	7	0
Tennessee	2	0
Texas	0	2
Washington	0	1
Wyoming	1	1
Total	58	28

*Reports to NFLIS-Drug are still pending for 2025 and 2026

Table 3: DHS Office of Field Operation on Drug Seizures at Port of Entry 2017-2019

	2017	2018	2019
<i>M.speciosa</i> (Mitragynine/7-hydroxymitragynine)	27.18 kg	0.52 kg	2,675.22 kg

FDA Warning Letters

Between June and July 2025, FDA issued seven warning letters⁸ to marketers and distributors for the unlawful use of 7-hydroxymitragynine as a drug, dietary supplements, or added to conventional food. The warning letters explicitly stated that 7-hydroxymitragynine is not an FDA-approved drug product and that no food additive regulation has authorized the use of 7-hydroxymitragynine in food. Instead, FDA classified 7-hydroxymitragynine as a “new dietary ingredient” under section 413(d) of the FD&C Act, 21 U.S.C. 350b(d)) because there is no evidence demonstrating it was marketed as a dietary ingredient in the United States before October 15, 1994. Some letters noted that as a new dietary supplement, 7-hydroxymitragynine is considered adulterated under section 402(f)(1)(B) of the FD&C Act, 21 U.S.C. 342(f)(1)(B)). This is because there is inadequate information providing reasonable assurance that the ingredient, 7-hydroxymitragynine, does not present a significant or unreasonable risk of illness or injury. Of note was FDA’s warning letter issued on June 25, 2025, to a company⁹ selling “7OHMZ 7-Hydroxymitragynine Gummies.” FDA cautioned that such products maybe appealing to children due to their packaging. Other warning letters targeted the illegal sale of unapproved 7-

⁸ [Warning Letters | FDA](https://www.fda.gov/news-events/press-announcements/fda-issues-warning-letters-firms-marketing-products-containing-7-hydroxymitragynine) [FDA Issues Warning Letters to Firms Marketing Products Containing 7-Hydroxymitragynine | FDA](https://www.fda.gov/news-events/press-announcements/fda-issues-warning-letters-firms-marketing-products-containing-7-hydroxymitragynine) <https://www.fda.gov/news-events/press-announcements/fda-issues-warning-letters-firms-marketing-products-containing-7-hydroxymitragynine> ; [7Tabz Retail, LLC - 709546 - 06/25/2025 | FDA](https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/7tabz-retail-llc-709546-06252025). <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/7tabz-retail-llc-709546-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/hydroxie-llc-709661-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/shaman-botanicals-llc-709622-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/royal-diamond-imports-inc-709540-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/thang-botanicals-inc-dba-7ohmz-7-ohmz-or-7ohmz-710190-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/relax-relief-rejuvenate-trading-llc-dba-rrr-trading-or-edp-kratom-709475-06252025>; <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/relax-relief-rejuvenate-trading-llc-dba-rrr-trading-or-edp-kratom-709475-06252025>.

⁹ Thang Botanicals, Inc. d/b/a 7OHMZ, 7-OHMZ, or 7OHMZMARCS-CMS 710190. June 25, 2025

hydroxymitragynine products marketed for the treatment of pain, relaxation, mood enhancement, and other medical conditions. Companies utilized websites and social media pages to make unproven medical claims for 7-hydroxymitragynine such as describing tablets as “expertly formulated to provide a potent dose” or promising “intense relaxation and a feeling of pure bliss.”

State Regulations and Controls

Due to concerns over its abuse, several states have regulated or banned the consumption of 7-hydroxymitragynine or *M. speciosa*.¹⁰ Currently, seven states (Alabama, Arkansas, Connecticut, Indiana, Louisiana, Vermont, and Wisconsin.) have prohibited 7-hydroxymitragynine consumption. Additionally, some states such as Arizona, Colorado, South Carolina, Texas have set restrictions on the limits of 7-hydroxymitragynine (not to exceed a specified percent of total alkaloid content). Furthermore, 19 states have enacted model legislation known as the “kratom consumer protection Act (KCPA),” which requires that *M.speciosa*, mitragynine, or 7-hydroxymitragynine be manufactured safely, labeled accurately, and distributed appropriately to protect consumers under a certain age. These states are Arizona, Colorado, Florida, Georgia, Kentucky, Oklahoma, Maryland, Mississippi, Nebraska, Nevada, New York, Oregon, Rhode Island, South Carolina, South Dakota, Texas, Utah, Virginia, and West Virginia. Other states like Illinois, New Hampshire, North Carolina, and Tennessee have bans in some localities.¹¹ Of note, the state of Mississippi has set the limit per weight basis to one percent of total alkaloid content of 7-hydroxymitragynine or 0.5 mg per container.

Poison Control Center Data (National Poison Data System)

The significance of 7-hydroxymitragynine abuse is demonstrated by an increasing volume of calls to poison control centers. Reporting for 7-hydroxymitragynine was historically limited within the national poison data system (NPDS); however, specific data codes for this substance were recently implemented between February and May 2025. During the initial tracking period from February 2025 to April 2025, NPDS recorded a total of 53 case reports involving 7-hydroxymitragynine. Of these exposure calls, 37 cases involved single-substance exposure of 7-hydroxymitragynine alone, and 24 cases were classified as abuse involving other substances in combination with 7-hydroxymitragynine. Among the single-substance exposure calls, 16 cases were categorized as intentional abuse, while 13 calls involved moderate medical outcomes where the patients exhibited pronounced and prolonged systemic symptoms. Expanded data from U.S poison centers¹² indicates a rapid escalation in reported incidents of 7-hydroxymitragynine. From January 1 through July 31, 2025, there were 165 exposure cases involving 7-hydroxymitragynine. Of those reporting exposures to 7-hydroxymitragynine alone, 35 percent of these cases resulted in serious health problems and 67 percent of individuals were treated at a healthcare facility. Patients exposed to 7-hydroxymitragynine frequently exhibit a range of severe physiological and neurological symptoms, including: gastrointestinal (nausea and

¹⁰ [Kratom Summary of State Laws](#). Accessed 08/08/2025

¹¹ [AKA In Your State](#). Accessed 02/26/2026

¹² [America's Poison Centers - Health Advisory: Serious Illnesses Associated with 7-OH Use](#). Accessed 08/26/2025

vomiting), neurological (agitation, confusion, loss of consciousness, and seizure), cardiovascular (sweating, tachycardia, and hypertension) and respiratory (difficulty breathing). According to HHS’s review, users report several reasons for using 7-hydroxymitragynine, including:

- **Desired Effects:** euphoria and an opioid-like “buzz”/high as motivation for using 7-hydroxymitragynine.
- **Product Appeal:** The availability of “candy-like” formulations of some 7-hydroxymitragynine tablets, which some users acknowledge as carrying health risk due to possibility of overconsumption.
- **Self-Treatment:** Claim of 7-hydroxymitragynine therapeutic value in self-treating pain and anxiety
- **Risk Awareness:** There is an acknowledgement among users of products containing 7-hydroxymitragynine that these products can lead to addiction, withdrawal symptoms, overdose and other serious health outcomes, including death (HHS review, 2025).

A study conducted a retrospective review of NPDS to identify additional exposures to 7-hydroxymitragynine that may not have been captured under its assigned product code from January 1, 2023, through July 31, 2025. The review showed that there were 186 exposure calls related to 7-hydroxymitragynine during that time period. Of those, 149 calls were associated with 7-hydroxymitragynine product code, and 37 exposure calls identified 7-hydroxymitragynine by substance verbatim.

In summary, the abuse of 7-hydroxymitragynine in the United States is fueled by its pharmacological similarities to opioid analgesics, a lack of regulatory controls, and the relative ease of obtaining 7-hydroxymitragynine products via smoke shops and the Internet.¹³ Furthermore, the consumption of 7-hydroxymitragynine along with other mind-altering substances may exacerbate the potential acute and long-term hazards and risks, like drug dependence, to the user. FDA responded to the increase in sales of 7-hydroxymitragynine products and unsubstantiated medical claims by issuing warning letters to companies to protect public safety. Available information from published literature and poison control centers suggests that 7-hydroxymitragynine is used by a diverse population for the self-treatment of various health conditions. The ingestion of 7-hydroxymitragynine, a potent opioid, alone or co-ingestion with other substances, commonly a CNS depressant, is of serious concern. The poison control center data and popularity of 7-hydroxymitragynine-products collectively underscores the severity and significance of abuse of 7-hydroxymitragynine in the United States.

Factor 6. What, if Any, Risk There Is to the Public Health

7-hydroxymitragynine has opioidergic and addictive properties. Available preclinical data demonstrates that 7-hydroxymitragynine has an abuse potential similar to that of schedule I and II opioids such as heroin, morphine, and fentanyl (Hemby *et al.*,

¹³ Internet sellers advertise, market, and provide false medical claims.

2019; Heustis *et al.*, 2024; Kruegel *et al.*, 2016; Matsumoto *et al.*, 2005; Obeng *et al.*, 2021). The abuse of 7-hydroxymitragynine presents severe risks to public health, including tolerance, dependence and addiction, respiratory depression, and death. Public health assessment of 7-hydroxymitragynine is further complicated because 7-hydroxymitragynine is a known metabolite of mitragynine, making it difficult to distinguish between the ingestion of *M. speciosa* and other isolated 7-hydroxymitragynine products.

As demonstrated below, data from programs that collect overdose or death events indicates that individuals are using product containing 7-hydroxymitragynine and experiencing harm.

FDA- Adverse Event Reporting Monitoring System (AEMS)

Data from AEMS shows an exponential increase in incidents involving 7-hydroxymitragynine. While only 3 cases were reported between 2023 and 2024, a DEA query on February 27, 2026, revealed that the 2025 case count had surged to 66. As of the most recent query on May 7, 2026, the cumulative total has reached 104 cases. The clinical severity of these reports is exceptionally high; 97 of the 104 cases are classified as serious, including 12 confirmed deaths. Data indicates that these adverse events are primarily driven by drug dependence and several withdrawal syndromes.

DEA Toxicology Testing Program (DEA TOX)

DEA TOX program, which investigates the presence of new psychoactive substances in biological samples from drug overdose, has identified 7-hydroxymitragynine in 85 cases since 2019. Of these, 55 were fatal and 30 were non-fatal cases. These cases involved both males and females with a median age of 36. The average concentration of 7-hydroxymitragynine detected in biological samples was 463.23 ng/mL. Samples often contained other related alkaloids (mitragynine and mitragynine pseudoindoxyl) or other drug classes, such as opioids (e.g. fentanyl), benzodiazepines (e.g. bromazolam), and ketamine. As shown below (see Table 4), between 2020 and 2026 there has been a significant increase in the positive identification of 7-hydroxymitragynine in fatal overdose cases.

Table 4: 7-hydroxymitragynine identification in DEA TOX Program (2019–present)

	Case Type	2019	2020	2021	2022	2023	2024	2025	2026*
7-hydroxymitragynine	Fatal	0	1	2	3	9	13	16	11
	Nonfatal	3	0	10	2	8	3	4	0
Total	Overdoses (fatal + nonfatal)	3	1	12	5	17	16	20	11

*2026 cases are still pending. As of March 2026.

Case Reports Involving 7-Hydroxymitragynine

In 2025, California,¹⁴ Pennsylvania,¹⁵ and Texas health authorities linked 7-hydroxymitragynine and concentrated extracts to several illness and overdose cases.¹⁶

Broul et al (2025) report follows a 31-year-old male who presented to the emergency department (ED) with catastrophic self-mutilation. The patient had used a sharp tool to amputate both of his ears and penis. He also presented with multiple lacerations on his forearms and frostbite on his toes. Toxicological analysis and patient history confirmed the presence of cannabinoids, mitragynine, morphine, and 7-hydroxymitragynine. Upon evaluation, the patient exhibited classic signs of acute substance-induced psychosis, including paranoia, psychomotor agitation, and inappropriate emotional responses (Broul et al., 2025).

Karinen et al (2014) reported a case of middle-aged decedent with a history of substance abuse and psychiatric disorder, who was found dead in his bed. Toxicological analysis conducted on the decedent revealed the presence of mitragynine (1.06 mg/L) and 7-hydroxymitragynine (0.15 mg/L) in the blood after a more comprehensive analysis. Routine toxicological screening detected the presence of zopiclone, citalopram, and lamotrigine in post-mortem whole blood. In the urine, mitragynine (3.47 mg/L) and 7-hydroxymitragynine (2.20 mg/L) were detected. The authors noted that due to 7-hydroxymitragynine's high opioid potency than mitragynine, it is likely it played a major part in the cause of death (Karinen et al, 2014).

Hendler et al (2026) published a case report involving a 30-year-old-male with opioid use disorder. He was hospitalized and treated with methadone 50 mg over 36-hour and later started on buprenorphine 16 mg/day.

Pullman et al (2026) report details the case of a 29-year-old male who suffered cardiopulmonary arrest after ingesting about 190 mg of 7-hydroxymitragynine tablets. The patient was found unresponsive and was revived with naloxone. The patient also disclosed habitual use where he consumed 90 mg daily, experiencing shaking and intense withdrawal symptoms only hours after his last dose (Pullman et al, 2026).

Wightman and Hu (2025) report on a case of a 38-year-old male with a history of opioid use disorder who transitioned from high-dose kratom capsules to over the counter 7-hydroxymitragynine tablets. Within a short period, his use escalated to about 240 mg/day. When he attempted to stop, he developed severe withdrawal syndrome including anxiety, insomnia, abdominal pain, and chills. His Clinical Opiate Withdrawal Scale score COWS

¹⁴ [LISTING OF DEPARTMENT OF PUBLIC HEALTH PRESS RELEASES.
http://publichealth.lacounty.gov/phcommon/public/media/mediapubhpdetail.cfm?prid=5156](https://publichealth.lacounty.gov/phcommon/public/media/mediapubhpdetail.cfm?prid=5156). October 10, 2025.

¹⁵ [Increased Volume of Calls Related to Kratom/Mitragynine and 7-hydroxymitragynine \(7-OH\) to Pennsylvania's Poison Centers.
https://www.pa.gov/content/dam/copapwp-pagov/en/health/documents/topics/documents/2025%20HAN/2025-802-%208-4-%20Kratom.pdf](https://www.pa.gov/content/dam/copapwp-pagov/en/health/documents/topics/documents/2025%20HAN/2025-802-%208-4-%20Kratom.pdf). August 4, 2025

¹⁶ [Serious Illnesses Associated with 7-OH Use | Texas DSHS; LISTING OF DEPARTMENT OF PUBLIC HEALTH PRESS RELEASES.
https://www.dshs.texas.gov/news-alerts/serious-illnesses-associated-7-oh-use](https://www.dshs.texas.gov/news-alerts/serious-illnesses-associated-7-oh-use). September 2, 2025

peaked at 14, indicating moderate-to-severe withdrawal comparable to classical opioids. Buprenorphine was initiated when his COWS exceeded 13. (Wightman and Hu, 2025).

These 7-hydroxymitragynine products are obtained through unknown sources, where the identity, purity, and concentration of active ingredients are often unknown, uncertain and inconsistent; thus, posing significant adverse health risks to users. Analyses of certain products have shown elevated concentrations of 7-hydroxymitragynine (Lydecker et al., 2016; CFSRE, 2025). Furthermore, some products are misleadingly marketed and sold as *M.speciosa*, leaving a user to unknowingly ingest a product containing a potent opioid, 7-hydroxymitragynine (Hill et al., 2025).

In summary, the abuse of 7-hydroxymitragynine poses a substantial hazard to public safety. 7-hydroxymitragynine is being abused for its opioid-like effects and shares health risks similar to other mu-opioid receptor agonists, such as morphine (schedule II). With no approved medical use, the positive identification of 7-hydroxymitragynine in non-fatal and fatal overdose cases poses threat to public safety. 7-Hydroxymitragynine products are obtained through unknown sources (commonly through the Internet); the identity, purity, and quantity of these substances are uncertain and inconsistent, thus posing significant adverse health risks to users.

Conclusion of 3-Factor Analysis

After a careful review of the scientific literature, scientific and medical evaluation and scheduling recommendation provided by HHS, Factors 4, 5, and 6, it is evident that the abuse of 7-hydroxymitragynine poses an imminent hazard to public safety. DEA has considered the three factors for temporary placement of a substance under schedule I of the CSA (21 U.S.C. 812). Available data indicate that 7-hydroxymitragynine above a specified threshold, including its isomers, esters, ethers, salts, and salts of isomers, esters and ethers, whenever the existence of such isomers, esters, ethers and salts is possible, poses an imminent hazard to public safety, has no currently accepted medical use in treatment in the United States, and lack accepted safety for use under medical supervision. Further, the misuse of this opioid poses a high risk for overdose and life-threatening events.

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